



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2005

MAY HALF YEARLY
EXAMINATION

Mathematics

(2 Unit Continuers)

General Instructions

- Write using black or blue pen.
- **Submit answers in 3 separate sections. Clearly marked A, B & C**
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Attempt all questions.

Total Marks = 100

Reading time - 5 minutes

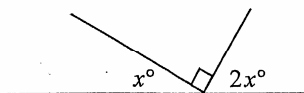
Working time - 90 minutes

Examiner: *P. Bigelow*

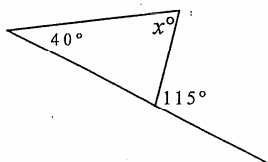
Section A (33 marks) – Start a new booklet**Marks**

- | | | |
|-----|---|---|
| 1. | Convert 40% to a fraction in its simplest form. | 1 |
| 2. | Find $\sin 16^\circ$ correct to 3 decimal places. | 1 |
| 3. | Simplify $2a^2 + 3a^2$. | 1 |
| 4. | Use a calculator to evaluate:
$\frac{11 \cdot 3 - 2 \cdot 6}{2 \cdot 4 + 3 \cdot 7}$ correct to 1 decimal place. | 1 |
| 5. | Express $\frac{4}{9}$ as a recurring decimal. | 1 |
| 6. | Simplify $\sqrt{12} + \sqrt{27}$. | 2 |
| 7. | Solve $5y + 1 = 2y + 7$. | 2 |
| 8. | Write down exact values for: | 2 |
| a.) | $\sin 60^\circ$ | |
| b.) | $\cos 225^\circ$ | |
| 9. | If $4\sqrt{5} = \sqrt{x}$, find x . | 1 |
| 10. | Write down the value of x in the following: | 3 |

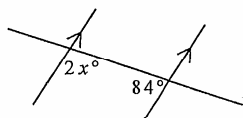
a.)



b.)



c.)



Section B (36 marks) – Start a new booklet**Marks**

- | | | |
|------------|---|---|
| 18. | Sketch $x - y + 4 = 0$ on a number plane. | 2 |
| 19. | If $f(x) = 2x^2 + 3x + 1$, find: | 2 |
| | a.) $f(0)$ | |
| | b.) $f(-3)$ | |
| 20. | What is the slope of the line $2x + y - 11 = 0$? | 1 |
| 21. | Find the angle sum of a regular decagon. | 2 |
| 22. | State the domain of the following functions: | 2 |
| | a.) $f(x) = \frac{1}{x-1}$ | |
| | b.) $f(x) = \sqrt{x-2}$ | |
| 23. | Expand and simplify: | 4 |
| | a.) $(4x+1)(x-2)$ | |
| | b.) $(\sqrt{5}-\sqrt{2})^2$ | |
| 24. | State whether the following functions are EVEN, ODD or NEITHER. | 3 |
| | a.) $f(x) = x^2 + 4$ | |
| | b.) $f(x) = x^2 - 3$ | |
| | c.) $f(x) = \frac{1}{1+x^2}$ | |
| 25. | Write down the complement of $26^\circ 37'$. | 1 |
| 26. | Solve: | 6 |
| | a.) $3a - 2(a-1) + 7 = 0$ | |
| | b.) $\frac{3x+4}{x} = 2$ | |
| | c.) $x^2 - 2x - 8 = 0$ | |

27. Sketch on separate number planes:

7

a.) $y = x^2$

b.) $y = 3^x$

c.) $xy = 6$

d.) $y = |x - 2|$

e.) $x^2 + y^2 \leq 9$

28. Find the equation of the line passing through the points (3,6) and (-1,4).

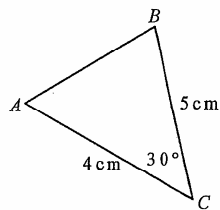
2

29. Express $\frac{3}{\sqrt{7}+2}$ with a rational denominator, in simplest form.

2

30. Find the area of $\triangle ABC$.

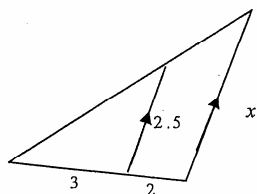
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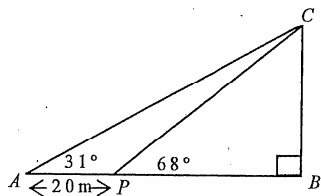
Section C (31 marks) – Start a new booklet

Marks

31. Factorise: 5
- a.) $y^2 - 25$
- b.) $x^2 - x - 6$
- c.) $3c^2 - 20c - 7$
- d.) $ab - a - bc + c$
- e.) $8 + a^3$
32. Solve the following pairs of simultaneous equations: 4
- a.) $x - y = 7$
 $x + y = 5$
- b.) $2x + 3y + 18 = 0$
 $x + 4y + 19 = 0$
33. Find a if $x^2 + 6x + a = (x + 3)^2$. 1
34. Solve $2x^2 - 6x - 9 = 0$, using the quadratic formula, (leave answers in simplest surd form). 2
35. Find the locus of points which are equidistant from the points $R(-4, 0)$ and $S(-2, 5)$. 3
36. Find the equation of the line passing through the intersection of $3x + y - 6 = 0$ and $2x - 3y + 4 = 0$, which contains the point $(5, -4)$. 3
37. Solve $\sin x = -\frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$. 2
38. Calculate the length of each side of a rhombus whose diagonals are 12cm and 16cm. 2
39. Find the value of x . 3



40. In $\triangle ABC$, $\angle B = 90^\circ$ and $\angle A = 31^\circ$. P is a point on AB such that $AP = 20\text{m}$, and $\angle CPB = 68^\circ$. 6



a.) Show that $PC = \frac{20 \sin 31^\circ}{\sin 37^\circ}$.

b.) Hence find PB correct to the nearest centimetre.

End of test.



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Sample Solutions

by
A. Ward

SECTION A

1) $\frac{2}{5}$

2) 0.276

3) $5a^2$

4) -1.4

5) $0.\dot{4}$

6) $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$

7) $3y = 6 \therefore y = 2$

8) a) $\sin 60^\circ = \sqrt{3}/2$

b) $\cos 225^\circ = -1/\sqrt{2}$

9) 80

10) a) $x = 30^\circ$

b) $x = 75^\circ$

c) $x = 48^\circ$

11) $x - 1 = 4 : x = 5$

$x - 1 = -4 : x = -3$

12) a) $m = 4$

b) $2\sqrt{17}$

c) $(3, -2)$

d) $x = 2$

13) $x = 4/33$

14) $a = -4$

15) $330 \times \frac{\pi}{180} = \frac{11\pi}{6}$

16) a) $\tan \theta = 3/4$

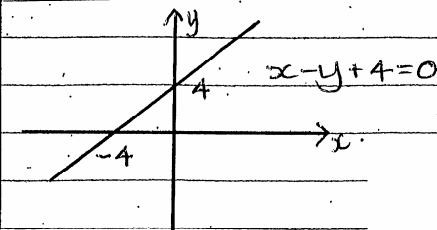
b) $\sec \theta = 5/4$

17) a) $b/12a$

b) $\frac{5x+1}{b}$

SECTION B

18)



28) $x - 2y + 9 = 0$

29) $\sqrt{7} - 2$

30) 5 cm^2

19) a) $f(0) = 1$

b) $f(-3) = -10$

20) $m = -2$

21) $(n-2)180 = 1440^\circ$

22) $x \in \mathbb{R} : x \neq 1$

$x \in \mathbb{R} : x \geq 2$

23) a) $4x^2 - 7x - 2$

b) $7 + 2\sqrt{10}$

24) a) EVEN

b) EVEN

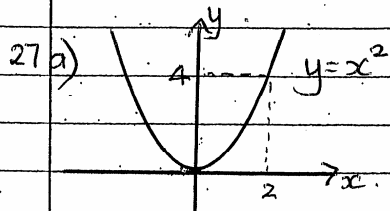
c) EVEN

25) $63^\circ 23'$

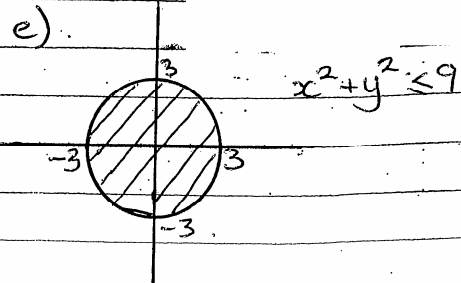
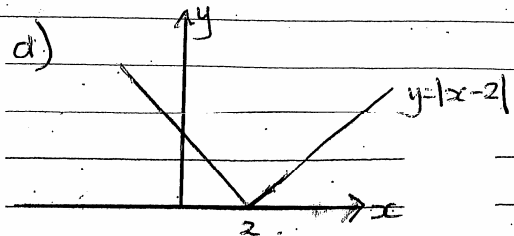
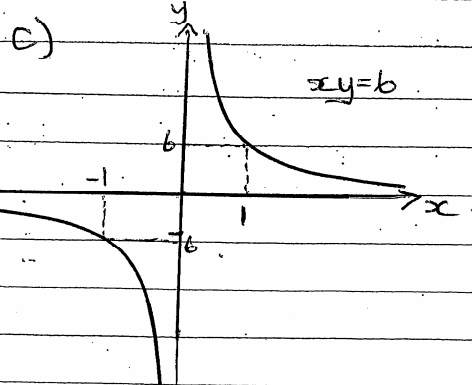
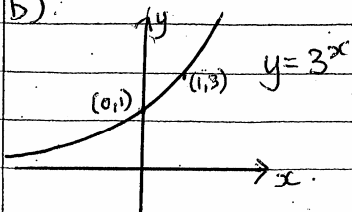
26) $a = -9$

$x = -4$

$x = 4 \text{ or } -2$



b)



SECTION C

31) a) $y^2 - 25 = (y+5)(y-5)$

b) $x^2 - x - 6 = (x-3)(x+2)$

c) $3c^2 - 20c - 7 = (3c+1)(c-7)$

d) $ab - a - bc + c = (a-c)(b-1)$

e) $8 + a^3 = (2+a)(4-2a+a^2)$

32) a) $x = 6$ $y = -1$

b) $y = -4$ $x = -3$

33) $a = 9$

34) $x = \frac{3 \pm 3\sqrt{3}}{2}$

35) $4x + 10y - 13 = 0$

36) $68x + 41y - 176 = 0$

37) b) $6 \cdot 412$ units

38) length of side $= \sqrt{6^2 + 8^2}$
 $= \sqrt{100} = 10 \text{ cm}$

39) $4\frac{1}{6}$